Final coefficient estimation

For each known order $p \in \{1, 2, 3, 4, 5\}$, 100 Jones coefficient vectors were sampled and reused across five Beta innovation shapes, giving 2500 series of length $T = 1000$. Innovations have population mean zero and variance $1/3$; 2000 burn-in observations were discarded [1, 2, 3].

The per-process coefficient error is

$$e_l(p) = \frac{1}{p} \sum_{j=1}^{p} \left| \widehat{\phi}_j^{(l)} - \phi_j^{(l)} \right|, \qquad l = 1, \ldots, 100.$$

Tables report the mean of $e_l(p)$ over the 100 processes in each condition [2]. All finite errors, including large errors and errors from nonstationary estimates, are included. Column minima are shown in bold. Table formatting follows Table 2.1 of the compiled thesis [4].

Table 1: Mean coefficient MAE with normalized $\mathrm{Beta}(1, 1)$ innovations.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.017736	0.018727	0.022171	0.023314	0.029758
Yule–Walker	0.017808	0.021261	0.030959	0.043539	0.052979
Horváth (L_∞)	**0.001617**	**0.002331**	**0.003414**	**0.004219**	**0.004741**
Minimum width	0.003381	0.014067	0.024996	0.053427	0.107356

Table 2: Mean coefficient MAE with normalized $\mathrm{Beta}(0.5, 0.5)$ innovations.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.017727	0.020761	0.023768	0.024429	0.028886
Yule–Walker	0.018447	0.022932	0.035126	0.042471	0.063128
Horváth (L_∞)	**0.000010**	**0.000020**	**0.000041**	**0.000056**	**0.000079**
Minimum width	0.000109	0.007798	0.017319	0.052198	0.087945

Table 3: Mean coefficient MAE with normalized $\mathrm{Beta}(2, 2)$ innovations.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.025190	**0.019233**	**0.024163**	**0.026980**	**0.029644**
Yule–Walker	0.024837	0.022302	0.028997	0.041771	0.057463
Horváth (L_∞)	**0.023048**	0.023585	0.028633	0.030559	0.034154
Minimum width	0.038506	0.058156	0.097299	0.124194	0.174185

Table 4: Mean coefficient MAE with normalized $\mathrm{Beta}(1, 2)$ innovations.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.020671	**0.018956**	**0.024503**	**0.025070**	**0.030801**
Yule–Walker	0.020980	0.020113	0.034665	0.038310	0.056849
Horváth (L_∞)	0.048276	0.047616	0.066165	0.066795	0.071414
Minimum width	**0.013709**	0.043746	0.060761	0.107466	0.155359

Table 5: Mean coefficient MAE with normalized $\mathrm{Beta}(2, 1)$ innovations.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.019560	**0.018090**	**0.023199**	**0.026013**	**0.028755**
Yule–Walker	0.019531	0.023971	0.029798	0.043879	0.068484
Horváth (L_∞)	0.051647	0.052771	0.057381	0.064830	0.070625
Minimum width	**0.018164**	0.035222	0.076280	0.094145	0.145433

Experiment details

The true order is supplied to every estimator. OLS and Horváth's LP use no intercept; Yule–Walker uses centered observations. Minimum width uses the fixed start e_1 and diameter-normalized steps

$$\gamma_m = \frac{0.1}{L}\, 2^{-m/100},$$

where L is the empirical point-cloud diameter. The cap is 2000 evaluated states, patience is 200, and the cumulative patience-reset threshold is $\delta = 10^{-8}$. Every strict width improvement is retained; no restarts or stationarity filtering are used [2].

All 10000 fits completed without fitting errors [1]. The companion `results.csv` contains the unrounded results; `provenance.json` records the source files and checksums.

References

[1] Saved final coefficient-estimation experiment, `accuracy_summary.csv` and `completion.json`, data set `1b5bb834ea4e8cc9`, run `eb96419f40fcab23`. A copy of the summary is provided as `results.csv`; `provenance.json` records the source files and checksums.

[2] Project implementations in `Code/Code_AR_processes/ar_tools/`: `final_coefficient_estimation.py`, `coefficient_errors.py`, and `innovations.py`. See `coefficient_mae`, `beta_innovation_bounds`, and the final estimation runner.

[3] M. C. Jones (1987). Randomly choosing parameters from the stationarity and invertibility region of autoregressive–moving average models. *Applied Statistics*, 36(2), 134–138. `https://doi.org/10.2307/2347544`.

[4] I. Sukhanov, compiled thesis, *Geometric Estimation of Autoregressive Processes and Spectral Scale Selection for Kernel Density Estimation*, Table 2.1, printed page 25.